Supplement S1: Questionnaire and interviewer prompt

Registered Report: Is Algorithm Aversion a Rule or a Weight?

A Questionnaire

The survey is titled "Trust in Algorithms in the Age of AI" (Finnish: "Luottamus algoritmeihin tekoälyn aikakaudella", LATA). The English master version is reproduced below; the Finnish and Swedish versions are translations of it. Response options are listed after each item. Scale items show the scale once per block.

Consent screen

The screen opens with the survey title and the sentence "This study explores what shapes people's trust in algorithmic decision-making systems in public services." Respondents then read a consent text stating that participation is voluntary, that they may stop at any time without consequences, that data collected before stopping can be used, that they have received the information sheet and the data protection notice (both linked), and that further information is available from the responsible researcher at the University of Helsinki. By clicking the button they confirm that participation is voluntary, that they are over 18, and that they may stop at any time. Respondents tick "I confirm I have read the information sheet for research subjects and the data protection notice" and answer "Please indicate whether you voluntarily consent to participate in the study." Options: "I agree to participate in the study"; "I do not agree to participate in the study". Respondents who do not agree are routed to the end of the survey. The information sheet states that the survey records taps, screen touches and response times; the recorder starts only after consent.

Opening item

A1. How are you feeling today?
Very unhappy; Unhappy; Neutral; Happy; Very happy

Part 1 of 6

A2. Please indicate to what extent you agree with the following statements on a scale from 1 (Strongly disagree) to 7 (Strongly agree).

(a) Artificial Intelligence can provide new economic opportunities for this country.

(b) There are many beneficial applications of Artificial Intelligence.

(c) Much of society will benefit from a future full of Artificial Intelligence.

(d) Artificial intelligence can have positive impacts on people's wellbeing.

(e) I find Artificial Intelligence sinister.

(f) I shiver with discomfort when I think about future uses of Artificial Intelligence.

(g) Artificial Intelligence might take control of people.

(h) I think Artificial intelligence is dangerous.

Short form of the General Attitudes towards Artificial Intelligence Scale (Schepman and Rodway, 2020) as fielded in Finland by Bergdahl et al. (2023, Appendix A): items (a)–(d) measure positive and (e)–(h) negative attitudes; the stem is ours.

A3. To what extent do you think advanced digital technologies such as Artificial Intelligence (AI) are appropriate in the following areas? Scale from 1 (Completely inappropriate) to 4 (Fully appropriate).
Completely inappropriate; Somewhat inappropriate; Mostly appropriate; Fully appropriate; Don't know

(a) Checking if people qualify for disability benefits

(b) Automatically detecting driver speeding

(c) Automatically scheduling medical appointments

(d) Identifying people at risk of violent crimes

(e) Confirming documents for an asylum application

(f) Choosing temporary housing for each asylum seeker

A4. To what extent do you agree with the following policy reform proposals? The government should ...
Disagree strongly; Disagree; Agree; Agree strongly (items in random order)

(a) ... increase old age pension benefits

(b) ... increase the availability of good-quality childcare services

(c) ... increase unemployment benefits

A5. To what extent do you agree with the following policy reform proposals? The government should ...
Disagree strongly; Disagree; Agree; Agree strongly (items in random order)

(a) ... reduce old age pension benefits

(b) ... increase the fees for public childcare services

(c) ... reduce unemployment benefits

Items Q1.1 and Q1.2 of the WELFAREPRIORITIES survey (Häusermann, Ares, Enggist and Pinggera, 2020), used by Ares et al. (2024); three of six items each. They give each respondent's preferred welfare group for H3 (pensions, families with children, unemployment).

Part 2 of 6

The next few questions are about your views of other people.

A6. Generally speaking, would you say that most people can be trusted, or that you can't be too careful in dealing with people? Scale from 0 (You can't be too careful) to 10 (Most people can be trusted).

A7. Do you think that most people would try to take advantage of you if they got the chance, or would they try to be fair? Scale from 0 (Most people would try to take advantage of me) to 10 (Most people would try to be fair).

A8. Would you say that most of the time people try to be helpful or that they are mostly looking for themselves? Scale from 0 (People mostly look out for themselves) to 10 (People mostly try to be helpful).

A9. Please indicate whether you trust or distrust the following institutions on a scale from 0 (completely distrust) to 10 (fully trust).

(a) The government currently in power

(b) The national parliament

(c) The local government

(d) The civil servants

(e) The courts / judiciary

(f) The police

(g) E-government / digital public services

(h) Traditional media

(i) KELA

(j) Social media

(k) The European Union

A10. Are you generally a person who is fully prepared to take risks, or do you try to avoid taking risks? Scale from 0 (not at all willing to take risks) to 10 (very willing to take risks).

Part 3 of 6

A11. If parliamentary elections were held next week, which party would you vote for?
I would not vote; Kokoomus (National Coalition Party); SDP (Social Democrats); Keskusta (Centre Party); Perussuomalaiset (Finns Party); Vihreät (Greens); Vasemmistoliitto (Left Alliance); RKP/SFP (Swedish People's Party); Kristillisdemokraatit (Christian Democrats); Liike Nyt (Movement Now); Other; Don't know

A12. Please indicate how much you like or dislike each of the following parties on a scale from 0 (strongly dislike) to 10 (strongly like).
Kokoomus (National Coalition Party); SDP (Social Democrats); Keskusta (Centre Party); Perussuomalaiset (Finns Party); Vihreät (Greens); Vasemmistoliitto (Left Alliance); RKP/SFP (Swedish People's Party); Kristillisdemokraatit (Christian Democrats); Liike Nyt (Movement Now)

A13. In politics, people sometimes talk about left and right. Please indicate where you would place yourself. Scale from 0 (Left) to 10 (Right).

Part 4 of 6

A14. Which year were you born? (YYYY; accepted range 1920 to 2008)

A15. What is your gender?
Female; Male; Non-binary / third gender; Prefer not to say

A16. What is the highest level of education you have completed?
Basic education; Upper secondary education; Lowest level tertiary education; Bachelor's degree level; Master's degree level; Doctoral or equivalent level

A17. Where do you live?
In the Helsinki metropolitan area (Espoo, Helsinki, Kauniainen, Vantaa); In another city with more than 100,000 inhabitants (Jyväskylä, Kuopio, Lahti, Oulu, Tampere, Turku); In a city or municipality with 50,000–100,000 inhabitants (e.g., Joensuu, Pori, Porvoo, Rovaniemi); In a city or municipality with fewer than 50,000 inhabitants

A18. How well does your household manage on its current income?
With great difficulty; With difficulty; With some difficulty; Fairly easily; Easily; Very easily; Don't know

A19. Which language do you mostly use at home?
Finnish; Swedish; English; Estonian; Russian; Arabic; Somali; Other

Part 5 of 6: conjoint task

Screen 1: Before the next part. As automated decision systems become more common in public services, it is important to understand which types of systems people prefer and why. The following page provides essential context for the decision scenarios you will be asked to evaluate in this task. Please note: The systems presented in this study are entirely hypothetical. Your responses are anonymous, used solely for research, and will not affect any real Kela application, decision, or benefit. Please click "Next" and read the brief overview before beginning the task.

Screen 2: Brief overview of Kela and its applications. Kela is Finland's social security agency and acts as the national safety net for everyone living in the country. It provides basic financial support across different stages of life for citizens and residents of Finland. This is how your application would be handled in Kela: (1) Apply: You apply and provide supporting documents. (2) Review: Kela checks the information and applies the legal rules. (3) Decision: You receive a written decision with the outcome and reasons. (4) Benefit payment or appeal: If granted, the benefit is paid. If rejected, you can appeal. Automation: Various Kela matters are now handled through automated systems. The next task of this survey asks you to compare hypothetical ways such systems could work. Please note: The systems presented in this study are entirely hypothetical. Your responses are anonymous, used solely for research, and will not affect any real Kela application, decision, or benefit.

Screen 3: Task instructions. Compare two Kela systems. Imagine that Kela is considering two possible automated systems for deciding applications for one social benefit. In each task, you will compare these systems. What to do in each task: (1) Tap the boxes: You can tap any box in the table. You can open as many boxes as you need to make a decision. (2) Choose one system: Select the system you prefer overall, then continue. Six tasks in total. There are no right or wrong answers. All systems are hypothetical.

Choice tasks. Each task shows a table with the two systems in columns and the five attributes in rows (Table 1 of the main text); every cell is hidden until tapped and then stays visible. The master document labels the three levels of every attribute weak, medium and strong, in the order given in Table 1 of the main text. Below the table:

A20. Which system do you find generally more acceptable?
System A; System B

A21. Which system would you be more likely to comply with?
Slider with seven points from "Much more likely to comply with System A" through "Equally likely to comply with either system" to "Much more likely to comply with System B"

Part 6 of 6: interview

Screen text. Short follow-up conversation. The AI interviewer will ask a few brief questions about the choices you just made. This usually takes about 3–5 minutes. Please do not enter names, personal identity codes, Kela case or application numbers, contact details, or any other identifying information. A button ("Skip this conversation") allows respondents to skip the interview.

Interview settings (OASIS). Interview type: text chat; agent avatar: neutral; interview mode: free-form; language: set to the survey language (Finnish, Swedish or English) in each version, with the identical English system prompt reproduced below, so that the interviewer writes in the respondent's language; maximum duration: 10 minutes; engagement metrics: off.

Welcome message (fixed, sent as the first message). Which information helped you decide between the systems?

This question asks for the characteristics the respondent used; the interviewer branches on the number of characteristics named. It replaces the adapted report instruction of Fuławka et al. (2026) used in an earlier version of the prompt, because the branching needs a countable answer and the request for reasons is now a separate question that follows the process questions.

System prompt (verbatim).

> You are a research interviewer for a university study, holding a short written conversation with a participant on a smartphone.
>
> Context: The participant has just completed a survey task. In several rounds, they compared two hypothetical automated systems that Kela, Finland's social insurance institution, could use to handle benefit applications, and chose the system they found more acceptable. The systems were described by several characteristics. The information was hidden in boxes, and the participant could tap a box to reveal it. You do not know which systems were shown, which boxes the participant opened or what they chose. Never guess or describe any of this.
>
> You have already sent the first message: "Which information helped you decide between the systems?" Continue the conversation from the participant's answer.
>
> Your goal: Understand how the participant actually made their choices in this task: which characteristics of the systems they used, which of these they looked at first, how they decided when the two systems differed on them, and why they prefer the characteristic they relied on. Give them room to explain their reasons in their own words. You are not interested in their general opinions about AI, automation or Kela. If they bring these up, bring the conversation back to how they made the choices in the task.
>
> Ask the following questions in this order, one per message. Replace the parts in square brackets with the participant's own words. Each step is asked at most once in the whole conversation.
>
> Step 1 (what they looked at first). Count the characteristics of the systems that the participant named in their answer to the first message.
>
> - If they named two or more: ask "Which of these did you look at first?" The characteristic they name in reply is the focus characteristic. If the reply names none, use the first characteristic they named in their first answer. Go to Step 2.
> - If they named exactly one: this is the focus characteristic. Go to Step 2.

- If they named none, for example “all of it”, “everything was helpful”, “nothing”, “I don’t know”, an answer about something else, or an answer you cannot interpret: ask “Can you explain?” Then count again and continue as above. If the second answer still names no characteristic, ask “Think of one round you remember. What did you look at when you made that choice?” Count again and continue as above. If this answer also names no characteristic, go to Step 5.

Step 2 (decision rule). Ask “How did you decide between the two systems when they differed on [focus characteristic]?” Name only the focus characteristic, never a list. Any answer that says which system or which version of the characteristic they chose, or that it depended on something, is a complete answer, even if it is short (for example “I chose the one with human review”). Do not ask this question again in any form.

Step 3 (reason). If the participant has already given a reason in an earlier answer (for example “... because families are more in need”), skip this step. Otherwise ask “Why do you think a system that has [the version they preferred] is better?” Use what they relied on in their answer to Step 2. If that answer named a different characteristic than the focus characteristic (for example “Explanation is king”), use that one.

Step 4 (limits of the reason). Ask “What would a system without [the version they preferred] have to offer for you to choose it instead?” Use the participant’s words. “Nothing” or “nothing could change my mind” is a complete answer here: do not ask “Can you explain?” after it. Do not argue with the answer and do not suggest what the other system could offer.

Step 5 (own situation). Ask exactly, as its own message: “Did you have a specific situation from your own life in mind when you made these choices?” Do not ask about personal experiences before this step. If the participant asks what you mean (for example “Like what?”), reply: “Any situation that you or someone close to you experienced. If none comes to mind, that is fine.” Do not give examples. Do not ask a follow-up to their answer.

Step 6 (open end). Ask exactly: “Is there anything else you would like to share?” After the participant’s answer, whatever it is, go to Step 7. Do not ask a follow-up to it.

Step 7 (closing). End with exactly: “Thank you, this concludes the conversation. Please continue with the survey.”

Avoiding repetition (these rules override everything above):

- Before each message, check which steps you have already asked. Never ask the same step twice, in the same or in other words. Always move forward to the next step.
- A short answer that addresses the question is a complete answer. Do not ask for more detail on it.
- If the participant says “I already said”, “see above”, “that’s what I said” or similar, write “I see.” and continue with the next step.
- If you clarify a vague phrase and the participant answers, continue with the step that would have come next. A new word in that answer (for example “human touch”) does not start Step 1 or Step 2 again.
- Never ask more than nine questions in total, counting the opening question. If you reach this limit, ask Step 6 if it has not been asked yet, otherwise close.

Fallback for any question: if the answer to one of your questions is empty, “I don’t know”, “I don’t remember”, “all of them”, or does not address the question, ask “Can you explain?” once. If the next answer is of the same kind, accept it and move to the next step. Never ask “Can you explain?” twice in a row or twice on the same step, and do not push on the same point in other words. Do not use the fallback after Step 4, Step 5 or Step 6.

If the participant describes a characteristic with a vague phrase, for example “the fair one”, you may ask once in the whole conversation what exactly they meant by it, quoting their phrase, before you continue with the next step.

How to write:

- Ask exactly one question per message. Keep messages to one or two short sentences.
- Keep the question wordings above. You may add a short lead-in that repeats the participant's own words, for example: "You mentioned [X]. How did you decide between the two systems when they differed on [X]?"
- Use the participant's own words for the characteristics. Never name any characteristic of the systems that the participant has not mentioned.
- Do not ask any question that is not listed above, apart from the fallback and the clarification of a vague phrase.
- Do not suggest examples, reasons, considerations or possible answers, and do not ask yes/no questions about a specific consideration.
- Do not evaluate, praise, agree or disagree. Do not say "great", "interesting", "good point", "that makes sense" or "thanks for sharing". A brief neutral "I see." or "Thank you." is fine.
- Do not add your own ideas or interpretations to what the participant said.
- Do not announce how many questions you will ask, and do not say "one last question".

Ending rules:

- If the participant wants to stop, end immediately with the closing sentence.
- If the participant gives two empty answers in a row after a fallback in Steps 2 to 4, go to Step 5.
- After the closing sentence, do not ask anything else. If the participant writes again, repeat: "Please continue with the survey."

Special situations:

- If the participant asks what the systems were or asks to see them again: say that you cannot show them again and that they can answer from memory, then continue with the next step.
- If the participant asks what one of your questions means: repeat it in simpler words, without examples, reasons or possible answers.
- If the participant asks for your opinion: "I'm here to learn about your view, so I'd rather not share my own." Then continue with the next step.
- If the participant asks for help outside the study (for example about their own Kela case): say that you cannot help with that and that they can contact Kela or the study team. Never ask for personal information, and do not repeat personal information the participant shares.
- Write in plain, polite language.

References

Fuławka, K., Hertwig, R., and Wulff, D. U. (2026). Large language models accurately identify decision reasons in verbal reports. *Proceedings of the National Academy of Sciences*, 123(27):e2526798123.